

Student Number: _____

Teacher: CJL HRK SKB

CRANBROOK SCHOOL

MATHEMATICS EXTENSION 1

2005

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

General Instructions

- Reading time: five minutes
- Working time: two hours
- Calculators may be used.

- The examination consists of 7 questions worth 12 marks each.
Begin a new booklet for each question
All questions should be attempted.
All necessary working should be shown in every question.
A table of standard integrals is provided at the back of this paper.

2

Question 1 (12 marks)

CJL

Marks

- (a) Let A be the point $(-3, 8)$ and let B be the point $(5, -6)$. Find the coordinates of the point P that divides the interval AB internally in the ratio $1:3$. 2
- (b) What is the remainder when the polynomial $P(x) = x^3 + 3x^2 - 1$ is divided by $x - 2$? 2
- (c) Use the table of standard integrals to find the exact value of 2
- $$\int_0^1 \frac{1}{\sqrt{x^2 + 9}} dx.$$
- (d) Solve $\frac{2}{x+5} \leq 1$. 3
- (e) Use the substitution $u = x - 1$ to evaluate $\int_2^4 \frac{x}{(x-1)^2} dx$. 3

Question 2 (12 marks)

CJL

Marks

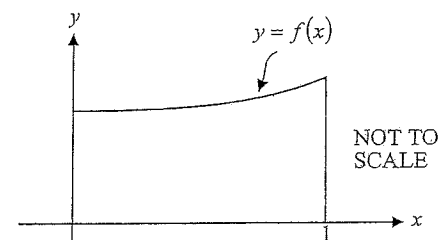
- (a) Sketch the graph of $y = 2\sin^{-1} 3x$ showing clearly the domain and range of the function as well as any intercepts. 2
- (b) Let $f(x) = 4x^2 - 1$. 2
Use the definition $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find the derivative of $f(x)$ at $x = a$.
- (c) Find $\frac{d}{dx}(3x^2 \cos^{-1} x)$. 2
- (d) Find $\int 4\cos^2 3x dx$. 2
- (e) Solve the equation $\sin 2\theta = \sqrt{2} \cos \theta$ for $0 \leq \theta \leq 2\pi$. 4

Question 3 (12 marks)

HRK

Marks

- (a) Find the acute angle between the lines $3x - 4y + 3 = 0$ and $y = 2x - 5$ to the nearest minute. 2
- (b) The function $f(x) = \log_e x + 5x$ has a zero near $x = 0.2$. Using $x = 0.2$ as a first approximation, use one application of Newton's method to find a second approximation to the zero. Write your answer correct to 3 decimal places. 3
- (c) (i) Find the natural domain of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$. 1
(ii) The sketch below shows part of the graph of $y = f(x)$. The area under the curve for $0 \leq x \leq 1$ is shaded. Find the area of the shaded region. 2



- (d) A particle moves in simple harmonic motion about a fixed point O . The amplitude of the motion is 2 m and the period is $\frac{2\pi}{3}$ seconds. Initially the particle moves from O with a positive velocity.
- (i) Explain why the displacement x , in metres, of the particle at time t seconds, is given by 1
$$x = 2\sin 3t$$
- (ii) Find the speed of the particle when it is $\sqrt{3}$ m from O . 2
- (iii) What is the maximum speed reached by the particle? 1

Question 4 (12 marks)

HRK

Marks

- (a) Use mathematical induction to prove that

3

$$1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1)$$

for all positive integers n .

- (b) The population N_t of Keystown first reached 25 000 on 1 January 2000. The population of Keystown is set to increase according to the equation

$$\frac{dN}{dt} = k(N - 8000)$$

where t represents time in years after the population first reached 25 000.

On 1 January 2005, the population of Keystown was 29 250.

- (i) Verify that $N = 8000 + Ae^{kt}$ is a solution to the above equation where A is a constant.
- (ii) Find the values of A and k .

1

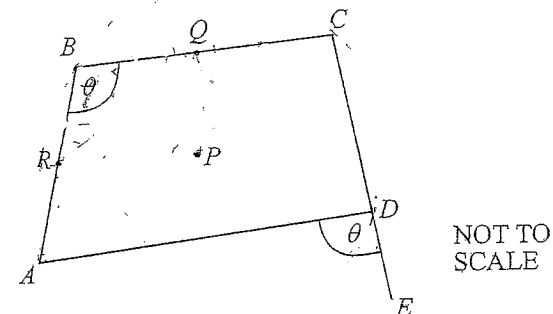
2

Question 4 continues on the next page.

Question 4 (continued)

Marks

(c)



In the diagram, R is the midpoint of AB , Q is the midpoint of BC and $AB = BC$. Point D lies on CE . Let $\angle ADE = \angle ABC = \theta$.

- (i) Explain why $ABCD$ is a cyclic quadrilateral.
- (ii) Given that P is the centre of the circle that passes through points A , B , C and D , show that $BQPR$ is a cyclic quadrilateral.
- (iii) Show that $\angle APR = \frac{180^\circ - \theta}{2}$.

1

2

3

Question 5 (12 marks)

HRK

Marks

- (a) If $\sin x = \frac{3}{5}$ and x is acute find the exact value(s) of $\tan \frac{x}{2}$.

3

- (b) Consider the function $f(x) = x(x-2a)$, $x \leq a$, where a is a constant

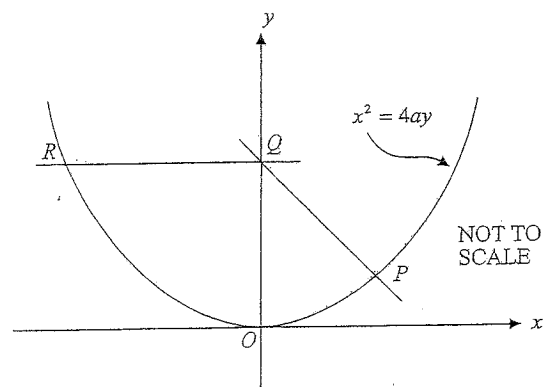
- (i) Find the values of a given that the inverse function $f^{-1}(x)$ of $f(x)$ exists.

2

- (ii) State the domain of $f^{-1}(x)$.

1

(c)



The diagram above shows the graph of the parabola $x^2 = 4ay$. The normal to the parabola at the variable point $P(2at, at^2)$, $t > 0$, cuts the y -axis at Q . Point R lies on the parabola.

- (i) Show that the equation of the normal to the parabola at P is $x + ty = at^3 + 2at$.
- (ii) Find the coordinates of R given that QR is parallel to the x -axis and $\angle PQR > 90^\circ$.
- (iii) Let M be the midpoint of RQ . Find the Cartesian equation of the locus of M .

2

2

2

Question 6 (12 marks)

SKB

Marks

- (a) A particle moves in a straight line with an acceleration given by

$$\frac{d^2x}{dt^2} = 9(x-2)$$

where x is the displacement in metres from an origin O after t seconds. Initially, the particle is 4 metres to the right of O so that $x = 4$ and has velocity $v = -6$.

- (i) Show that $v^2 = 9(x-2)^2$.

2

- (ii) Find an expression for v and hence find x as a function of t .

2

- (iii) Explain whether the velocity of the particle is ever zero.

2

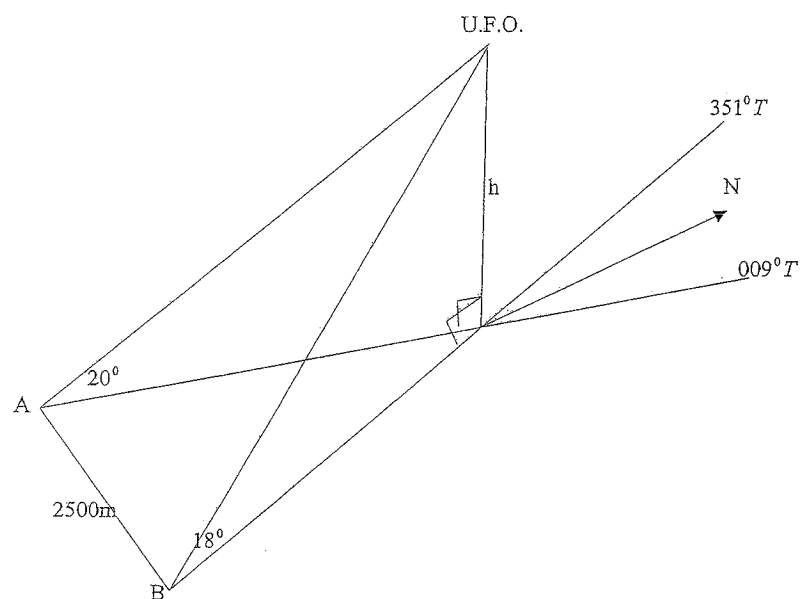
Question 6 continues on the next page.

Question 6 (continued)

Marks

(b) Two surveyors A and B standing 2500 metres apart on level ground take the bearing and elevation of a U.F.O. at the same instant. Surveyor A notes the bearing as $009^\circ T$ and the angle of elevation as 20° , whilst Surveyor B finds the corresponding bearing and elevation to be $351^\circ T$ and 18° respectively. If the U.F.O. is h metres above the ground:

- (i) Prove that $h = \frac{2500}{\sqrt{\cot^2 20^\circ + \cot^2 18^\circ - 2 \cot 20^\circ \cot 18^\circ \cos 18^\circ}}$. 4
- (ii) Hence evaluate the height, h to the nearest metre. 2



Question 7 (12 marks)

SKB

Marks

- (a) (i) Write $8 \sin \theta - 15 \cos \theta$ in the form $R \sin(\theta - \alpha)$ where α is acute. 1
- (ii) Hence or otherwise find the general solutions of $8 \sin \theta - 15 \cos \theta = \frac{-17\sqrt{3}}{2}$ to the nearest minute. 3
- (b) The roots of the equation $x^3 - 12x^2 + 12x + m = 0$ are in arithmetic progression. Find the numerical value of m and the roots of the equation. 4
- (c) Evaluate $\int_0^3 \frac{dx}{9+x^2}$ by using the substitution $x = 3 \tan \theta$. 4



**2005 CRANBROOK
MATHEMATICS EXTENSION 1
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION SOLUTIONS**

Question 1 (12 marks)

(a) $P = \left(\frac{lx_1 + kx_2}{k+l}, \frac{ly_1 + ky_2}{k+l} \right)$

where $k=1$, $l=3$, $x_1=-3$, $y_1=8$, $x_2=5$ and $y_2=-6$.

$$\text{So } P = \left(\frac{3 \times -3 + 1 \times 5}{4}, \frac{3 \times 8 + 1 \times -6}{4} \right) \\ = \left(-1, 4\frac{1}{2} \right)$$

2 marks	Correct answer
1 mark	Either the x or the y coordinate of P correct

(b) $P(x) = x^3 + 3x^2 - 1$

Using the Remainder Theorem,

$$P(2) = 8 + 12 - 1$$

$$= 19$$

The remainder is 19.

2 marks	Correct answer
1 mark	Correct method with an arithmetic mistake

(c) From the table of standard integrals, we have

$$\int_0^1 \frac{1}{\sqrt{x^2+9}} dx = \left[\ln(x + \sqrt{x^2+9}) \right]_0^1 \\ = \ln(1 + \sqrt{10}) - \ln(0+3) \\ = \ln \frac{1+\sqrt{10}}{3}$$

2 marks	Correct answer
1 mark	Correct first line above

Question 1 (continued)

(d) Method 1 Multiplying both sides by $(x+5)^2$

$$\frac{2}{x+5} \leq 1, \quad x \neq -5$$

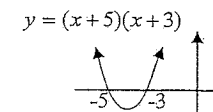
$$2(x+5) \leq (x+5)^2$$

$$2x+10 \leq x^2+10x+25$$

$$0 \leq x^2+8x+15$$

$$0 \leq (x+5)(x+3)$$

From the graph of $y = (x+5)(x+3)$ we see that $y \geq 0$ for $x \leq -5$ or $x \geq -3$ but $x \neq -5$ from above so $x < -5$ or $x \geq -3$.



3 marks	Correct answer
2 marks	Correct method but with one mistake
1 mark	Finds either of the critical points $x = -5$ or $x = -3$ or states that $x \neq -5$

Method 2 Critical points method

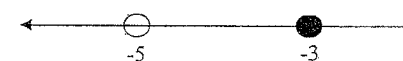
$$\frac{2}{x+5} \leq 1, \quad x \neq -5$$

Consider $\frac{2}{x+5} = 1$

$$2 = x+5$$

$$x = -3$$

Look at the number line.



For $x = -6$, $\frac{2}{-6+5} \leq 1$

For $x = -4$, $\frac{2}{-4+5} \geq 1$

For $x = -1$, $\frac{2}{-1+5} \leq 1$

So $\frac{2}{x+5} \leq 1$ for $x \leq -5$ or $x \geq -3$, but $x \neq -5$

So $x < -5$ or $x \geq -3$

3 marks	Correct answer
2 marks	Correct method but with one mistake
1 mark	Finds either of the critical points $x = -5$ or $x = -3$ or states that

$x \neq -5$

Question 1 (continued)

Method 3 Consider the cases where $x+5 > 0$ and $x+5 < 0$

$$\frac{2}{x+5} \leq 1, \quad x \neq -5$$

Now, $2 \leq x+5$ if $x+5 > 0$

$$x \geq -3 \text{ if } x > -5$$

So $x \geq -3$

Also, $2 \geq x+5$ if $x+5 < 0$

$$x \leq -3 \text{ if } x < -5$$

So $x < -5$

So $x < -5$ or $x \geq -3$

3 marks	Correct answer
2 marks	Correct method but with one mistake
1 mark	Finds either of the critical points $x = -5$ or $x = -3$ or states that $x \neq -5$

(e)

$$\int_2^4 \frac{x}{(x-1)^2} dx = \int_1^3 (u+1)u^{-2} \frac{du}{dx} dx$$

where $u = x-1$,

$$= \int_1^3 (u^{-1} + u^{-2}) du$$

$$\frac{du}{dx} = 1,$$

and $x = u+1$

$$= [\log_e u - u^{-1}]_1^3$$

Also, $x = 4$ so $u = 3$

$$= \left(\log_e 3 - \frac{1}{3} \right) - (\log_e 1 - 1)$$

and $x = 2$ so $u = 1$

$$= \log_e 3 + \frac{2}{3} \quad (\text{since } \log_e 1 = 0)$$

3 marks	Correct answer
2 marks	Finds correct integrand and terminals OR Follows correct method but forgets to change terminals
1 mark	Finds correct terminals OR finds correct integrand

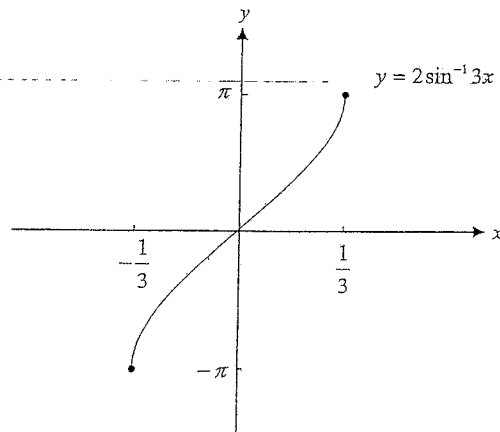
Question 2 (12 marks)

- (a) The function
- $y = 2\sin^{-1} 3x$
- is defined for

$$\begin{aligned} -1 &\leq 3x \leq 1 \\ -\frac{1}{3} &\leq x \leq \frac{1}{3} \end{aligned}$$

The coefficient 2 has the effect of stretching vertically the basic graph of $y = \sin^{-1} 3x$ by a factor of 2.

Since the graph of $y = \sin^{-1} 3x$ has a range of $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, the graph of $y = 2\sin^{-1} 3x$ has a range of $-\pi \leq y \leq \pi$.



2 marks	Showing correct graph with endpoints clearly marked
1 mark	Correct shape of graph without clearly marked endpoints

Question 2 (continued)

(b) $f(x) = 4x^2 - 1$

$f(a) = 4a^2 - 1$

$$\begin{aligned} f(a+h) &= 4(a+h)^2 - 1 \\ &= 4a^2 + 8ah + 4h^2 - 1 \end{aligned}$$

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4a^2 + 8ah + 4h^2 - 1 - 4a^2 + 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{8ah + 4h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(8a + 4h)}{h} \\ &= \lim_{h \rightarrow 0} (8a + 4h) \\ &= 8a \end{aligned}$$

2 marks	Correct derivation
1 mark	Correct substitution into definition

$$\begin{aligned} \text{(c)} \quad \frac{d}{dx} (3x^2 \cos^{-1} x) &= 6x \cos^{-1} x + 3x^2 \times \frac{-1}{\sqrt{1-x^2}} \quad (\text{Product rule}) \\ &= 6x \cos^{-1} x - \frac{3x^2}{\sqrt{1-x^2}} \end{aligned}$$

2 marks	Correct answer
1 mark	Reasonable attempt to use product rule OR Showing $\frac{d}{dx} (\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$

$$\begin{aligned} \text{(d)} \quad \int 4 \cos^2 3x \, dx &= 4 \int \left(\frac{1}{2} + \frac{1}{2} \cos 6x \right) dx \quad \text{since } \cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta \\ &= 4 \left(\frac{x}{2} + \frac{1}{12} \sin 6x \right) + c \\ &= 2x + \frac{1}{3} \sin 6x + c \end{aligned}$$

2 marks	Correct answer
1 mark	Correct substitution of identity for $\cos^2 3x$

Question 2 (continued)

(e) $\sin 2\theta = \sqrt{2} \cos \theta$

$2 \sin \theta \cos \theta = \sqrt{2} \cos \theta$

$\sqrt{2} \sin \theta \cos \theta - \cos \theta = 0$

$\cos \theta (\sqrt{2} \sin \theta - 1) = 0$

$\cos \theta = 0$

or

$\sin \theta = \frac{1}{\sqrt{2}}$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$

S	A
T	C

4 marks	Four correct answers
3 marks	Correct factorisation and finding $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ OR Correct factorisation and finding $\theta = \frac{\pi}{4}, \frac{3\pi}{4}$
2 marks	Correct factorisation
1 mark	Correct replacement of $\sin 2\theta$

Question 3 (12 marks)

(a) For $3x - 4y + 3 = 0$ $m_1 = \frac{3}{4}$ and for $y = 2x - 5$ $m_2 = 2$.

Now for acute angle θ between lines: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$\therefore \tan \theta = \left| \frac{\frac{3}{4} - 2}{1 + \frac{3}{4} \times 2} \right| \therefore \tan \theta = \frac{1}{2} \therefore \theta = 26^\circ 34' \text{ (to nearest minute).}$

2 marks	Correct answer
1 mark	Determining correct gradients and stating correct formula.

Question 3 (continued)

(b) Now, $f(x) = \log_e x + 5x$

$f'(x) = \frac{1}{x} + 5$

and $x_0 = 0.2$

Using Newton's method

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$= 0.2 - \frac{\log_e 0.2 + 5 \times 0.2}{\frac{1}{0.2} + 5}$

$= 0.261 \text{ (correct to 3 decimal places)}$

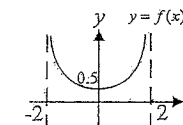
3 marks	Correct answer
2 marks	Finds $f(0.2)$ and $f'(0.2)$ correctly and substitutes into formula
1 mark	Correctly evaluates $f'(0.2)$

(c) (i) Method 1Because of the square root sign, $4 - x^2 \geq 0$ so $-2 \leq x \leq 2$.Also however, because $\sqrt{4 - x^2} \neq 0$, $x \neq -2, 2$.So the natural domain of f is $-2 < x < 2$.

1 mark	Correctly reasoned answer
--------	---------------------------

Method 2

Sketch the graph.

The natural domain of f is $-2 < x < 2$.

1 mark	Correct answer and correct graph
--------	----------------------------------

Question 3 (continued)

$$\begin{aligned}
 \text{(ii) Area} &= \int_0^1 \frac{1}{\sqrt{4-x^2}} dx \\
 &= \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_0^1 \\
 &= \sin^{-1} \frac{1}{2} - \sin^{-1} 0 \\
 &= \frac{\pi}{6}
 \end{aligned}$$

Area required is $\frac{\pi}{6}$ square units.

2 marks	Correct answer
1 mark	Correct first line above

- (d) (i) The particle starts from the centre of motion and moves with positive velocity so the general form of the displacement time function is $x = a \sin nt$.

$$\text{Now } a = 2 \text{ and period} = \frac{2\pi}{n}$$

$$\text{so } n = 3$$

So the required equation is $x = 2 \sin 3t$ as required.

1 mark	Correctly derived equation
--------	----------------------------

Question 3 (continued)

$$\text{(ii) } x = 2 \sin 3t$$

$$\text{When } x = \sqrt{3},$$

$$\sin 3t = \frac{\sqrt{3}}{2}$$

$$3t = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \dots$$

$$t = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{7\pi}{9}, \dots$$

$$\text{Now } x = 2 \sin 3t$$

$$\frac{dx}{dt} = 6 \cos 3t$$

$$\text{When } t = \frac{\pi}{9},$$

$$\frac{dx}{dt} = 6 \cos \frac{\pi}{3}$$

$$= 3$$

$$(\text{Check when } t = \frac{2\pi}{9},$$

$$\frac{dx}{dt} = 6 \cos \frac{2\pi}{3}$$

$$= 6 \times -\frac{1}{2},$$

$$= -3$$

$$\text{speed} = |-3|$$

$$= 3)$$

So speed is 3 m s^{-1} .

S	A
T	C

S	A
T	C

2 marks	Correct answer
1 mark	Finding correct values of t when particle is $\sqrt{3}$ m from O OR Substituting incorrect value of t into correct expression for $\frac{dx}{dt}$ OR Giving an answer of -3 m s^{-1}

Question 3 (continued)

(iii) Method 1For SHM the maximum speed occurs at the centre of motion i.e. at O .

$$x = 2 \sin 3t$$

$$0 = 2 \sin 3t$$

$$3t = 0, \pi, 2\pi, \dots$$

$$t = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \dots$$

$$\frac{dx}{dt} = 6 \cos 3t$$

When $t = 0$,

$$\begin{aligned} \frac{dx}{dt} &= 6 \cos 0 \\ &= 6 \end{aligned}$$

$$\begin{array}{c|c} S & A \\ \hline T & C \end{array}$$

(Check when $t = \frac{\pi}{3}$,

$$\frac{dx}{dt} = 6 \cos \pi$$

$$= -6$$

$$\begin{aligned} \text{Speed} &= |-6| \\ &= 6 \end{aligned}$$

Maximum speed reached by particle is 6 m s^{-1} .

1 mark	Correct answer
--------	----------------

Method 2Maximum speed occurs when $\frac{d^2x}{dt^2} = 0$

$$\frac{d^2x}{dt^2} = -18 \sin 3t$$

$$0 = -18 \sin 3t$$

$$3t = 0, \pi, 2\pi, \dots$$

$$t = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \dots$$

$$\frac{dx}{dt} = 6 \cos 3t$$

When $t = 0$,

$$\begin{aligned} \frac{dx}{dt} &= 6 \cos 0 \\ &= 6 \end{aligned}$$

(Check when $t = \frac{\pi}{3}$,

$$\frac{dx}{dt} = 6 \cos \pi$$

$$= -6$$

$$\begin{aligned} \text{Speed} &= |-6| \\ &= 6 \end{aligned}$$

Maximum speed reached by particle is 6 m s^{-1} .

1 mark	Correct answer
--------	----------------

Question 4 (12 marks)

- (a) Prove that $1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1)$ using mathematical induction

$$\begin{aligned} \text{When } n=1, \quad RHS &= \frac{1}{6} \times 3 \times 2 \\ &= 1 \\ &= LHS \end{aligned}$$

So the statement is true for $n=1$.Suppose that it is true for $n=k$.

$$\text{That is, suppose that } 1 + 6 + 15 + \dots + k(2k-1) = \frac{1}{6}k(4k-1)(k+1) \quad - (A)$$

Then, when $n=k+1$, we have

$$\begin{aligned} 1 + 6 + 15 + \dots + (k+1)(2(k+1)-1) &= \frac{1}{6}(k+1)(4(k+1)-1)((k+1)+1) \\ &= \frac{1}{6}(k+1)(4k+3)(k+2) \end{aligned}$$

$$\begin{aligned} LHS &= 1 + 6 + 15 + \dots + (k+1)(2(k+1)-1) \\ &= 1 + 6 + 15 + \dots + k(2k-1) + (k+1)(2(k+1)-1) \\ &= \frac{1}{6}k(4k-1)(k+1) + (k+1)(2k+1) \quad \text{from (A)} \\ &= (k+1) \left\{ \frac{1}{6}k(4k-1) + (2k+1) \right\} \\ &= (k+1) \left\{ \frac{1}{6}k(4k-1) + \frac{1}{6}(12k+6) \right\} \\ &= \frac{1}{6}(k+1)(4k^2 - k + 12k + 6) \\ &= \frac{1}{6}(k+1)(4k+3)(k+2) \\ &= RHS \end{aligned}$$

It is true for $n=1$ and when it is true for $n=k$ it has been proven that it is true for $n=k+1$. Hence by mathematical induction it must be true for all positive integers n .

3 marks	Shows correct proof
2 marks	Shows proof for $n=1$ AND shows correct assumption statement for $n=k$ and substitutes $n=k+1$ correctly into the expression
1 mark	One of either of the points mentioned immediately above

Question 4 (continued)

(b) (i) Consider $\frac{dN}{dt} = k(N - 8000)$

$$LS = \frac{dN}{dt}$$

$$= \frac{d}{dt}(8000 + Ae^{kt})$$

$$= Ake^{kt}$$

$$RS = k(N - 8000)$$

$$= k(8000 + Ae^{kt} - 8000)$$

$$= Ake^{kt}$$

$$= LS$$

Have verified.

1 mark	Shows that $N = 8000 + Ae^{kt}$ satisfies the differential equation OR Finds the general solution to the differential equation
--------	---

(ii) When $t = 0$, $N = 25000$

$$\text{So, } 25000 = 8000 + Ae^0$$

$$A = 17000$$

When $t = 5$, $N = 29250$

$$\text{So, } 29250 = 8000 + 17000e^{5k}$$

$$21250 = 17000e^{5k}$$

$$1.25 = e^{5k}$$

$$5k = \log_e 1.25$$

$$k = 0.2 \log_e 1.25$$

So $A = 17000$ and $k = 0.2 \log_e 1.25$

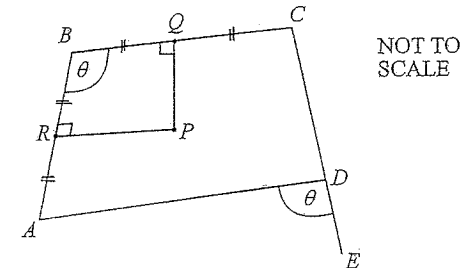
2 marks	Correct answers for A and k
1 mark	Correct answer for A OR Correct answer for k

Question 4 (continued)

(c) (i) $\angle ADE = \angle ABC = \theta$ and hence $ABCD$ is a cyclic quadrilateral because the exterior angle at a vertex of a cyclic quadrilateral equals the interior opposite angle.

1 mark Correct answer

(ii)



Since P is the centre of the circle that passes through the points A , B , C and D , then QP is the perpendicular bisector of BC and PR is the perpendicular bisector of AB .

(The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord.)

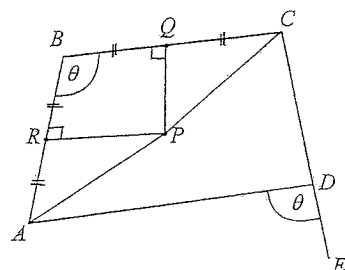
So $\angle BQP = \angle BRP = 90^\circ$

So $BQPR$ is a cyclic quadrilateral. (The opposite sides of a cyclic quadrilateral are supplementary).

2 marks	Correct reasoning
1 mark	Stating with reasons that QP and PR are the perpendicular bisectors of BC and AB respectively.

Question 4 (continued)

(iii)



NOT TO SCALE

Now, $\angle QPR = 180^\circ - \theta$ (Opposite sides of a cyclic quadrilateral are supplementary.)

and $\angle APC = 2\theta$ (A, B and C lie on a circle and the angle at the centre of a circle is twice the angle at the circumference subtended by the same arc.)

Also, $PQ = PR$ (Equal chords are equidistant from the centre of the circle.)

$CQ = AR$ (Since $AB = BC$ and R and Q are respective midpoints.)

$AP = CP$ (Radii of a circle are equal.)

So $\triangle APR \equiv \triangle CPQ$ (Three pairs of sides are equal in length.)

So $\angle CPQ = \angle APR$

Now, $\angle QPR + \angle CPQ + \angle CPA + \angle APR = 360^\circ$

So, $180^\circ - \theta + \angle APR + 2\theta + \angle APR = 360^\circ$

$$\angle APR = \frac{180^\circ - \theta}{2}$$

as required.

3 marks	Correct reasoning including showing $\triangle APR \equiv \triangle CPQ$
2 marks	Showing congruence correctly but incorrectly reasoning the rest of the case
1 mark	Showing correct reasoning without showing $\triangle APR \equiv \triangle CPQ$

Question 5 (12 marks)

(a) If $t = \tan \frac{x}{2}$, $x \neq \pi$ then $\sin x = \frac{2t}{1+t^2}$.

$$\therefore \frac{3}{5} = \frac{2t}{1+t^2}$$

$$\therefore 3t^2 - 10t + 3 = 0$$

$$\therefore (3t-1)(t-3) = 0$$

$$\therefore t = \frac{1}{3} \text{ or } 3$$

$$\therefore \tan \frac{x}{2} = \frac{1}{3} \text{ or } 3$$

But as x is acute $\tan \frac{x}{2} \neq 3 \therefore \tan \frac{x}{2} = \frac{1}{3}$ only.

3 marks	Correct derivation of answer
2 marks	Gives both answers for $\tan \frac{x}{2}$
1 mark	Uses a correct t result as part of the solution

Question 5 (continued)

- (b) (i) The inverse function $f^{-1}(x)$ exists if the graph of $y = f(x)$ is 1:1.

From the graph we see that $f(x)$ is not a 1:1 function. That is, a horizontal line can be drawn that will cut the graph at more than one point.

One of the turning points is $(2, 0)$. Find the other turning point.

$$y = x(x-2)^2 \\ = x^3 - 4x^2 + 4x$$

$$\frac{dy}{dx} = 3x^2 - 8x + 4$$

$$= (3x-2)(x-2)$$

$$\text{So, } (3x-2)(x-2) = 0$$

$$x = \frac{2}{3} \text{ or } x = 2$$

So $f^{-1}(x)$ exists if $a \leq \frac{2}{3}$.

2 marks	Correct answer
1 mark	Giving answer $a \leq \frac{2}{3}$ OR Giving an incorrect inequality as an answer after an arithmetic mistake in the differentiation or solution to the quadratic

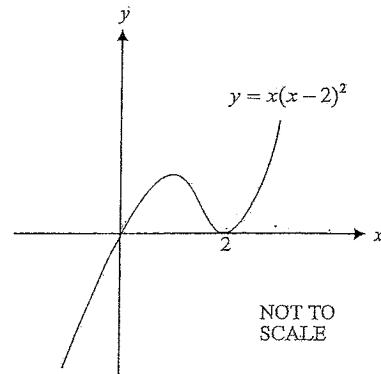
- (ii) The domain of $f(x)$ is $x \leq \frac{2}{3}$.

The range of $f(x)$ is $y \leq \frac{32}{27}$.

$$\left(\text{Since } f\left(\frac{2}{3}\right) = \frac{2}{3}\left(\frac{2}{3} - 2\right)^2 = \frac{32}{27}\right)$$

So the domain of $f^{-1}(x)$ is $x \leq \frac{32}{27}$.

1 mark	Correct answer
--------	----------------



Question 5 (continued)

- (c) (i) Method 1 – Using the Cartesian equation

$$x^2 = 4ay$$

$$2x = 4a \frac{dy}{dx} \quad (\text{implicit differentiation})$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$\text{At } P(2at, at^2),$$

$$\frac{dy}{dx} = \frac{2at}{2a}$$

$$= t$$

So the gradient of the normal at P is $-\frac{1}{t}$.

Equation of the normal at P :

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$ty - at^3 = -x + 2at$$

$$x + ty = at^3 + 2at \text{ as required}$$

2 marks	Derives correct gradient of normal AND Derives correct equation of normal
1 mark	Derives correct gradient of normal

Method 2 – Using parametric equations

$$\text{At } P(2at, at^2),$$

$$x = 2at \text{ and } y = at^2$$

$$\frac{dx}{dt} = 2a \quad \frac{dy}{dt} = 2at$$

$$\text{So } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \quad (\text{Chain rule})$$

$$= 2at \cdot \frac{1}{2a}$$

$$= t$$

So the gradient of the normal at P is $-\frac{1}{t}$.

Equation of the normal at P :

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$ty - at^3 = -x + 2at$$

$$x + ty = at^3 + 2at \text{ as required}$$

2 marks	Derives correct gradient of normal AND Derives correct equation of normal
1 mark	Derives correct gradient of normal

Question 5 (continued)

- (ii) The normal at P passes through Q which lies on the y -axis.
 Now, $x + ty = at^3 + 2at$,
 when $x = 0$, $y = at^2 + 2a$
 So Q and therefore R have a y -coordinate of $y = at^2 + 2a$.
 Since R lies on the parabola, substitute $y = at^2 + 2a$ into $x^2 = 4ay$.

$$\begin{aligned} x^2 &= 4a(at^2 + 2a) \\ &= 4a^2t^2 + 8a^2 \\ &= 4a^2(t^2 + 2) \end{aligned}$$

Since $\angle PQR > 90^\circ$

$$x = -2a\sqrt{t^2 + 2}$$

If $\angle PQR < 90^\circ$, $x = 2a\sqrt{t^2 + 2}$

So R has coordinates $(-2a\sqrt{t^2 + 2}, at^2 + 2a)$

2 marks	Finding two correct coordinates
1 mark	Finding one correct coordinate

$$(iii) \quad M = \left(\frac{-2a\sqrt{t^2 + 2} + 0}{2}, \frac{at^2 + 2a + at^2 + 2a}{2} \right)$$

$$= \left(-a\sqrt{t^2 + 2}, at^2 + 2a \right)$$

$$x = -a\sqrt{t^2 + 2}$$

$$y = at^2 + 2a$$

$$\text{So } x = -a\sqrt{\frac{y - 2a}{a} + 2}$$

$$= -a\sqrt{\frac{y - 2a + 2a}{a}}$$

$$x = -a\sqrt{\frac{y}{a}} \text{ is the Cartesian equation of the locus of } M.$$

($x = -\sqrt{ay}$ is also an acceptable equation)

2 marks	Correct equation
1 mark	Correct coordinates of M

Question 6 (12 marks)

$$(a) \quad (i) \quad \frac{d^2x}{dt^2} = 9(x - 2)$$

$$\frac{d\left(\frac{1}{2}v^2\right)}{dx} = 9(x - 2)$$

$$\frac{1}{2}v^2 = 9 \int (x - 2) dx$$

$$= 9\left(\frac{x^2}{2} - 2x\right) + c$$

Initially $x = 4$ and $v = -6$

$$\text{So, } 18 = 9(8 - 8) + c$$

$$c = 18$$

$$\frac{1}{2}v^2 = 9\left(\frac{x^2}{2} - 2x\right) + 18$$

$$v^2 = 9x^2 - 36x + 36$$

$$= 9(x^2 - 4x + 4)$$

$$= 9(x - 2)^2$$

as required

2 marks	Correct integration AND Correct evaluation of c
1 mark	Statement that $\frac{d\left(\frac{1}{2}v^2\right)}{dx} = 9(x - 2)$

Question 6 (continued)

(ii) From (i), $v^2 = 9(x-2)^2$
 So $v = \pm 3(x-2)$, but
 when $t = 0$, $x = 4$ and $v = -6$.
 So, $v = -3(x-2)$
 So, $\frac{dx}{dt} = -3(x-2)$
 $\frac{dt}{dx} = \frac{-1}{3(x-2)}$
 $t = -\frac{1}{3} \int \frac{1}{x-2} dx$
 $= -\frac{1}{3} \log_e(x-2) + c$
 $t = 0$, $x = 4$
 $0 = -\frac{1}{3} \log_e 2 + c$
 $t = \frac{1}{3} \log_e \frac{2}{x-2}$
 $3t = \log_e \frac{2}{x-2}$
 $e^{3t} = \frac{2}{x-2}$
 $x-2 = 2e^{-3t}$
 $x = 2(1 + e^{-3t})$

2 marks	Correct expression for $x(t)$
1 mark	Correct expression for $v(x)$

(iii) From (ii), $v = -3(x-2)$
 If $v = 0$, $0 = -3(x-2)$
 So $x = 2$
 From (ii) also, $x = 2(1 + e^{-3t})$
 If $x = 2$, $2 = 2(1 + e^{-3t})$
 $e^{-3t} = 0$ which has no solution.

So v is never zero.

Alternatively, the graph of $x = 2(1 + e^{-3t})$ has an asymptote of $x = 2$ and so $x \neq 2$ so $v \neq 0$.

2 marks	Correct explanation involving $v(x)$ and $x(t)$
1 mark	Correct explanation only as far as involving $v(x)$

Question 6 (continued)

- (b) (i) Let O be at the foot of the UFO to the ground.
 Let OB = y and OA = x .

$$\therefore \tan 18^\circ = \frac{h}{y} \therefore y = h \cot 18^\circ \dots\dots\dots 1.$$

$$\therefore \tan 20^\circ = \frac{h}{x} \therefore x = h \cot 20^\circ \dots\dots\dots 2.$$

$$\therefore 2500^2 = x^2 + y^2 - 2xy \cos 18^\circ \dots\dots\dots 3.$$

Sub. 1 and 2 into 3:

$$\therefore 2500^2 = h^2 \cot^2 20^\circ + h^2 \cot^2 18^\circ - 2h \cot 20^\circ \times h \cot 18^\circ \cos 18^\circ$$

$$\therefore 2500^2 = h^2 (\cot^2 20^\circ + \cot^2 18^\circ - 2 \cot 20^\circ \cot 18^\circ \cos 18^\circ)$$

$$\therefore h = \frac{2500}{\sqrt{\cot^2 20^\circ + \cot^2 18^\circ - 2 \cot 20^\circ \cot 18^\circ \cos 18^\circ}}$$

4 marks	Correct derivation.
3 marks	Derivation to second last equation.
2 marks	Listing of the three equations needed in derivation.
1 mark	Listing of at least two of the equations needed in the derivation.

(ii) $h = 2583.015192\dots\dots$

$$\therefore \text{height, } h = 2583 \text{ m (to nearest metre)}$$

2 marks	Finds correct answer
1 mark	Does not round off correctly to nearest metre.

Question 7 (12 marks)

(a) (i) $8 \sin \theta - 15 \cos \theta = R \left[\frac{8}{R} \sin \theta - \frac{15}{R} \cos \theta \right]$

where $R = \sqrt{8^2 + (-15)^2} = 17$.

$\therefore 8 \sin \theta - 15 \cos \theta = 17 \sin(\theta - \alpha)$

where $\cos \alpha = \frac{8}{17}, \sin \alpha = \frac{15}{17}$

such that $\tan \alpha = \frac{15}{8}$ and $\alpha = \tan^{-1} \frac{15}{8}$ for acute α .

$\therefore 8 \sin \theta - 15 \cos \theta = 17 \sin(\theta - \tan^{-1} \frac{15}{8})$.

1 mark	Correct derivation
--------	--------------------

(ii) If $8 \sin \theta - 15 \cos \theta = \frac{-17\sqrt{3}}{2}$

then $17 \sin(\theta - \tan^{-1} \frac{15}{8}) = \frac{-17\sqrt{3}}{2}$

$\therefore \sin(\theta - \tan^{-1} \frac{15}{8}) = \frac{-\sqrt{3}}{2}$

$\sin(\theta - \tan^{-1} \frac{15}{8}) = \sin(-60^\circ)$

$\theta - \tan^{-1} \frac{15}{8} = 180^\circ n + (-1)^n (-60^\circ)$

$\theta = 180^\circ n + (-1)^n (-60^\circ) + 61^\circ 56'$ to the nearest minute
where n is any integer.

3 marks	Correct derivation
2 marks	Correctly finding θ but not qualifying what n is.
1 mark	Correct to 3 rd last line of working.

(b)

Let the roots of $x^3 - 12x^2 + 12x + m = 0$ be $a-d, a, a+d$.

Now sum of roots $= a-d + a + a+d$

$= 3a$

$= 12$

$\therefore a = 4$

$\Rightarrow x = 4$ is a root and $P(4) = 0$ where $P(x) = x^3 - 12x^2 + 12x + m$.

$\therefore 0 = 64 - 192 + 48 + m$

$0 = -80 + m$

$\therefore m = 80$

Now $P(x) = x^3 - 12x^2 + 12x + 80$

and the product of the roots $= (a-d) a (a+d)$

$= a(a^2 - d^2)$

$= 4(16 - d^2)$

$= -80$

$\therefore 16 - d^2 = -20$

$\therefore d^2 = 36$

$\therefore d = \pm 6$

$\Rightarrow$ roots are $-2, 4$ and 10 .

(c)

$I = \int_0^3 \frac{dx}{9+x^2}$

letting $x = 3 \tan \theta \quad \therefore \frac{dx}{d\theta} = 3 \sec^2 \theta$

when $x = 0 \quad \theta = 0$

$x = 3 \quad \theta = \frac{\pi}{4}$

$\therefore I = \int_0^{\frac{\pi}{4}} \frac{3 \sec^2 \theta \, d\theta}{9 + 9 \tan^2 \theta}$
 $= \int_0^{\frac{\pi}{4}} \frac{3 \sec^2 \theta \, d\theta}{9(1 + \tan^2 \theta)}$
 $= \int_0^{\frac{\pi}{4}} \frac{3 \sec^2 \theta \, d\theta}{9 \sec^2 \theta}$
 $= \int_0^{\frac{\pi}{4}} \frac{1}{3} \, d\theta$

$= \left[\frac{\theta}{3} \right]_0^{\frac{\pi}{4}}$

$= \frac{\pi}{12}$

4 marks	Correct derivation
3 marks	Correctly derived to third last line of working.
2 marks	Correctly derived to fifth last line of working.
1 mark	Correctly finding $\frac{dx}{d\theta}$ and changing the values of the limits with respect to θ .